



Review article

Behavioral shifts in finance through the lens of prospect theory: A systematic literature review and bibliometric analysis

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ABSTRACT

The study aims to provide an objective and systematic review of the applications of prospect theory (PT) in finance. To do that, we undertake a hybrid review combining a systematic literature review, bibliometric analysis using VOSviewer and Biblioshiny, and content analysis. In bibliometric analysis, we find that PT research spans both decision science and mainstream finance journals. The research is concentrated in the United States, China, and Western Europe, with emerging economies existing on the periphery. The author collaboration network is fragmented, which indicates a lack of coordinated research development. Further, in content analysis, we note that the recent literature has increasingly focused on the cross-sectional applications of PT. At the same time, the growth in foundational themes such as the equity premium puzzle and the endowment effect has been sluggish. Moreover, we unravel conflicting findings regarding the stability of the loss-aversion coefficient and the sign of the relationship between returns and PT across markets, while also noting that the empirical base for bonds and cryptocurrencies remains narrow. The study makes several contributions, including the first hybrid review of PT in finance and the first cross-asset synthesis covering equities, mutual funds, bonds, and cryptocurrencies.

1. Introduction

How investors assess risk and uncertainty forms the core of finance research. Expected Utility Theory (EUT, henceforth) has long served as the workhorse for modeling decisions under risk (Schoemaker, 1982; Tversky and Kahneman, 1974). EUT underpins the seminal models of modern finance, including Modern Portfolio Theory, the Capital Asset Pricing Model (CAPM), and the Fama-French three-factor model (Fama and French, 1992; Lintner, 1965; Markowitz, 1952; Sharpe, 1964). Yet a now-large body of empirical work documents systematic departures of investor behavior from EUT predictions. These inconsistencies pose substantial difficulties in explaining various stylized facts in stock returns (Barberis and Huang, 2008; Boyer et al., 2010; Brunnermeier et al., 2007), leading researchers to seek new approaches that better reflect the complexities of how investors make choices.

Prospect Theory (PT, henceforth), proposed by Kahneman and Tversky (1979) and refined as Cumulative Prospect Theory by Tversky and Kahneman (1992), has emerged as one of the most widely deployed alternatives. Integrating key behavioral finance concepts, such as reference-dependent risk preferences, loss aversion, and decision

weights, PT has been used to rationalize various interesting phenomena, including the disposition effect, the endowment effect, lottery preferences, and the equity premium puzzle (EPP, henceforth). These mechanisms have also been used to construct a measure of PT value that predicts the cross-section of equity, mutual-fund, bond, and cryptocurrency (CC, henceforth) returns (Barberis et al., 2016; Barberis and Huang, 2008; Benartzi and Thaler, 1995; Chen et al., 2022b; Zhong and Wang, 2018). Moreover, in behavioral finance, which is no longer on the periphery of financial literature, PT is its central theoretical anchor.

The volume and reach of PT-related research in finance have grown tremendously over the past decade. The applications have been extended to new asset classes (mutual funds and CCs) and new geographies. However, this growth has not been uniform. While cross-sectional asset-pricing applications of PT have proliferated, foundational themes such as the EPP have grown more slowly. Further, several empirical claims rest on a limited set of replications and untested assumptions about parameter stability, return distributions, and arbitrage limits. A comprehensive synthesis is therefore both timely and necessary.

While several studies have examined PT's applications in finance, a

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Table 1
EUT vs. PT.

Dimension	Expected Utility Theory (EUT)	Prospect Theory (PT)
Carrier of value	Final states of wealth, including existing assets	Changes in wealth (gains and losses) are defined relative to a reference point
Shape of the value/utility function	Concave over wealth, implying diminishing marginal utility	S-shaped value function: concave for gains, convex for losses, and steeper for losses than for gains
Risk attitude	Risk aversion over the entire outcome domain	Risk aversion for gains and risk seeking for losses
Treatment of losses	Losses and gains of equal magnitude are treated symmetrically; no separate loss-aversion parameter.	Losses loom larger than gains; loss aversion captured by parameter λ ($\lambda = 2.25$)
Probability weighting	Outcomes weighted linearly by their objective probabilities	Decision weights $w(p)$ are a non-linear transformation of objective probabilities; small probabilities are overweighted, and moderate-to-high probabilities are underweighted.
Invariance to framing	Preferences are invariant to how outcomes are described (invariance axiom)	Preferences depend on how outcomes are coded as gains or losses, called the framing effect

Notes: Table constructed from the seminal works on EUT and PT (Kahneman and Tversky, 1979; Tversky and Kahneman, 1981, 1992; Von Neumann and Morgenstern, 1944)

comprehensive systematic literature review (SLR, henceforth) and a bibliometric analysis are long overdue.¹ In the corpus used for this review, 72% of the articles were published after 2013. The last comprehensive review of PT by Barberis (2013) predates the recent surge in PT-finance research and the integration of PT into formal cross-sectional asset-pricing tests. Earlier reviews (De Giorgi and Hens, 2006; Edwards, 1996; Han and Hsu, 2004; Jose and Mutuc, 2010) are narrative in nature, predate the use of PT-value as a significant cross-sectional predictor, and mostly cover a literature dominated by laboratory evidence rather than the application in the financial market. More recent reviews have been limited in scope. For instance, Ogunmokun et al. (2023) examine PT only in the context of bank credit decisions, and Saliya (2020) summarizes PT vis-à-vis EUT. Recent bibliometric reviews of loss aversion in investment (Soman et al., 2026) and of the disposition effect (Gutiérrez-Nieto et al., 2023) focus on a single PT component or mechanism. Dhingra et al. (2024) examine behavioral biases broadly rather than PT specifically. Therefore, aggregating and synthesizing the current body of knowledge, along with identifying gaps in the literature, are of utmost importance for advancing research on PT in financial markets. To do so, we conduct a hybrid review that integrates (i) an SLR following Tranfield et al. (2003) and the SPAR-4-SLR² protocol of Paul et al. (2021), (ii) a bibliometric mapping using VOSviewer (van Eck and Waltman, 2010) and Biblioshiny in RStudio (Aria and Cuccurullo, 2017), and (iii) content analysis of the most influential studies.

We rely on the Scopus database for our literature search.³ The hybrid design is deliberate. Whereas SLR helps select and evaluate the extant literature in a more transparent and reproducible manner (Tranfield et al., 2003), and bibliometric analysis enables the evaluation of large

numbers of published scientific articles, enabling a more comprehensive review (Bhaskar et al., 2022; Donthu et al., 2021). The bibliometric analysis is conducted on 218 articles ranked ABDC A*, A, and B to capture the full intellectual structure of the field. Whereas keeping journal quality as the proxy for methodological rigor and influence on subsequent research (Currie and Pandher, 2011; Engemann and Wall, 2009), content analysis has been restricted to the 126 ABDC A*/A articles. Further, to navigate the literature systematically, we organize our literature synthesis along the natural hierarchy of PT, i.e., (i) theoretical components (the value function and the probability-weighting function), (ii) behavioral mechanisms (loss aversion, the disposition and endowment effects, and lottery-like preferences), and (iii) empirical applications (cross-sectional return prediction, the EPP).

Thus, the research objective (RO, henceforth) of the study is fourfold: 1) to consolidate and map the expanding academic literature on PT; 2) to identify key trends and recent developments in the field; 3) to facilitate future research by pinpointing gaps in the literature; and 4) to offer valuable insights for investors, academics, researchers, and regulators.

The bibliometric analysis reveals that PT research is published in both decision-science and mainstream finance journals. The top fourteen journal outlets account for about 40% of the corpus. While the co-citation structure is tightly consolidated around Tversky & Kahneman's foundational work, the empirical work applying PT to financial markets is dispersed. Studies are concentrated in the US, China, and Western Europe, with emerging-market and culture-specific evidence sitting at the edges of the network.

Content analysis unravels several important observations. We find that there are conflicting results on the magnitude and stability of the loss-aversion coefficient (Blavatsky, 2023; Ert and Erev, 2013) and on the sign of the return-PTV relationship across markets and investor types (Barberis et al., 2016; do Nascimento Junior et al., 2021; Son and Nguyen, 2019). Further, the explanatory power of any single PT component in isolation is limited. Loss aversion alone does not reconcile the EPP, and the disposition effect arises from PT only under restrictive parameter configurations in which diminishing sensitivity dominates loss aversion. Though the cross-sectional applications of PT extend well beyond equities and the bond market, CC evidence currently has a very narrow empirical base (Chen et al., 2022b; Yadav, 2024; Zhong and Wang, 2018).

This study makes several key contributions. First, it is the first hybrid review of PT in finance, combining SLR, bibliometric mapping, and content analysis. Second, the study provides the first comprehensive cross-asset synthesis of PT applications, covering equities, mutual funds, bonds, and CCs. Third, the study moves beyond descriptive documentation and provides a critical analysis of the literature by identifying conflicting empirical findings and highlighting key debates. Finally, it concludes with a concrete research agenda for future researchers.

Beyond this, the review also carries important implications for investors, fund managers, and market regulators. For investors and fund managers, the cross-asset evidence on the return-PTV relationship explains where PT-driven mispricing is most likely to persist and where arbitrage forces are likely to dominate. For regulators, documented behavioral patterns, such as the disposition effect among retail investors and the strengthening of lottery demand during market downturns, may inform the framework for investor protection rules and financial literacy interventions in markets.

The structure of the paper is as follows. Section 2 presents an overview of the PT model. Section 3 outlines the methodology. Section 4 discusses the bibliometric results. Section 5 presents the content analysis of major themes from the literature. Section 6 details the limitations, research gaps, and future research agenda. Section 7 concludes the paper.

¹ While the focus of this review is limited to PT applications in finance, Camerer (1998) has very useful discussions of PT applications in various domains of economics.

² Stands for "Scientific Procedures and Rationales for Systematic Literature Reviews"

³ Robustness check using WOS is reported in Appendix A

2. A note on PT and PT model

2.1. EUT vs PT

The widespread acceptance and application of EUT as both a descriptive model of economic behavior and a normative model of rational choice suggests that rational individuals should, and generally do, follow its axioms. Both EUT and PT recognize that the utility individuals derive from wealth differs from its monetary value, focusing on the satisfaction individuals obtain from their current wealth or changes therein. Furthermore, both theories predict that investors exhibit risk aversion in the gains domain, and an increase in wealth yields diminishing marginal utility. However, PT differs from EUT on several key dimensions, which are summarized in Table 1.

2.2. PT model

Original PT, proposed by Kahneman and Tversky (1979), though it had all the essentials of a theory, had some limitations. For instance, the original PT applied only to gambles with at most two outcomes, and people sometimes chose dominated gambles. However, these principles limit the practical application of the theory. In 1992, a subsequent paper expanded the framework and resolved the issues with the original model, introducing Cumulative Prospect Theory⁴ (Tversky and Kahneman, 1992). This refined model (a) accommodates prospects with any number of outcomes, (b) addresses both risk and uncertainty, and (c) differentiates between the evaluation of gains and losses. PT's comprehensive framework has made it the go-to version for economic analysis; a brief overview is given below.

PT breaks down decision-making into two phases. The initial phase, the editing stage, involves individuals forming a mental framework for their decision prospects. In the valuation phase, value is assigned to these prospects, and choices are made. During the editing stage, individuals simplify decision prospects by transforming their representations through mental operations such as coding, simplification, dominance detection, combination, segregation, and cancellation of common components. Kahneman and Tversky (1979, pp. 274–275) discuss these editing operations in greater detail. They argue that formulating editing theory is difficult as it is influenced by the "norms, habits, and expectancies of the decision maker." In other words, a formal algorithm does not govern the editing stage; rather, it reflects the decision maker's prior experiences, established heuristics, and social and cultural expectations, all of which influence how a prospect is mentally framed before it is evaluated. Subsequently, each prospect undergoes an assessment in the second phase. The one that emerges with the highest value is selected. The overall value of a prospect is decomposed into two parts: value function [$v(x_i)$] and a PW function [$w(p_i)$]. Consider a gamble with the following distribution:

$$(x_{-m}, p_{-m}; x_{-m+1}, p_{-m+1}; \dots; x_0, p_0; \dots; x_{n-1}, p_{n-1}; x_n, p_n)$$

Here x_{-m} is the extreme negative outcome, with p_{-m} probability and x_n is the extreme positive outcome with p_n probability. Outcomes here are arranged in ascending order.

Following Tversky and Kahneman (1992, p.309), the value of a prospect is defined as:

$$PTV = \sum_{i=-m}^n \pi_i v(x_i) \quad (1)$$

Here, PTV is prospect theory value, $v(\cdot)$ is the "value function" and π_i represents the "probability-weighting function". The value function, as proposed, takes the following form:

$$v(x_i) = \begin{cases} x_i^\alpha & \text{if } x_i \geq 0 \\ -\lambda(-x_i)^\alpha & \text{if } x_i < 0 \end{cases} \quad (2)$$

For gains ($x \geq 0$), $v(x) = x^\alpha$ and for losses ($x < 0$), $v(x) = -\lambda(-x)^\alpha$. The value function is defined by three key features: (1) It is based on deviations from a reference point (gains and losses) rather than the final level of wealth as in EUT. Therefore, if the reference point changes, the value function shifts accordingly. (2) It is generally concave for gains and convex for losses, indicating risk aversion for gains and risk-seeking behavior for losses. (3) It is steeper for losses than for gains, with losses having roughly twice the impact of gains, according to experimental evidence from Tversky and Kahneman (1992). Parameter λ captures individuals' loss aversion. Parameter α shapes the S-curve of the value function, reflecting the principle of diminishing sensitivity. This principle suggests that the impact of a change weakens with the distance from the reference point. For instance, in the loss space, a fall from \$15 to \$10 has greater psychological value than a fall from \$105 to \$100. The parameter estimates reported by Tversky and Kahneman (1992, p.311–312), derived from experimental data, are $\alpha = 0.88$, $\lambda = 2.25$, $\gamma = 0.61$, and $\delta = 0.69$.

The second component of the PTV is a probability-weighting function (π_i) that captures the decision weights. Following Tversky and Kahneman (1992, p.301), it is represented as:

$$\begin{aligned} \pi_n^+ &= w^+(p_n), \quad \pi_m^- = w^-(p_{-m}), \\ \pi_n^+ &= w^+(p_i + \dots + p_n) - w^+(p_{i+1} + \dots + p_n), \quad 0 \leq i \leq n-1 \\ \pi_m^- &= w^-(p_{-m} + \dots + p_i) - w^-(p_{-m} + \dots + p_{i-1}), \quad 1-m \leq i < 0 \end{aligned} \quad (3)$$

The decision weights are calculated using a weighting function, usually denoted as $w(\cdot)$. This function takes objective probability (P) as its input, where the specific functional forms of $w^+(p)$ and $w^-(p)$ are given by Tversky and Kahneman (1992, p.309).

where,

$$w^+(p) = \frac{p^\gamma}{(p^\gamma + (1-p)^\gamma)^{\frac{1}{\gamma}}}, \quad w^-(p) = \frac{p^\delta}{(p^\delta + (1-p)^\delta)^{\frac{1}{\delta}}} \quad (4)$$

In Eq. (3) π_i represents the PW function. In Eq. (4), $w^-(p)$ and $w^+(p)$ are the decision weights assigned to negative and positive outcomes, respectively. In the probability-weighting function, probabilities are distorted near the endpoints. Highly unlikely events are given more weight than outcomes having moderate to high probabilities (reflecting the possibility effect), capturing the lottery preferences of the decision-makers. At the other end, the certainty effect prevails, with improbable outcomes overweighted relative to those with moderate to high probability. This distortion at the endpoints shows the individuals' simultaneous demand for lottery and insurance. Individuals favor a 0.001 chance of \$100,000 to a certain gain of \$1000 but also prefer a certain loss of \$1000 to a 0.001 chance of losing \$100,000. This behavior is not consistent with the predictions of the EUT. The smaller the parameter values (γ, δ), the greater the overweighting of extreme outcomes. Further, following Tversky and Kahneman (1992), several studies have validated the value and weighting parameters of the PT (Abdellaoui, 2000; Gonzalez and Wu, 1999).

3. Methodology

3.1. Method

This study employs a hybrid literature review approach, integrating SLR with bibliometric and content analysis (Brahmi et al., 2023, 2024; Cheng et al., 2016; Tranfield et al., 2003). In contrast to conventional literature reviews that summarize and evaluate other researchers' work (Wakefield, 2014), an SLR ensures a well-defined search protocol,

⁴ In this study, we use the term "PT" to refer to cumulative prospect theory (CPT), while the term "original PT" is used specifically to denote the initial version of prospect theory.

research questions (RQs, henceforth), data extraction, and presentation (Ahmad et al., 2020; Kitchenham et al., 2009). This approach is better in accommodating large bibliographic datasets and ensuring objectivity, comprehensiveness, and replication in the methodology.⁵ Subsequently, building upon SLR, bibliometric analysis provides an overview of the subject area (Bhaskar et al., 2022; Donthu et al., 2021), and in this study, we attempt to offer an overview of the current research landscape on the application of PT in finance (Donthu et al., 2021; Li et al., 2017). Bibliometric analysis offers several benefits, including efficient processing of massive bibliographic datasets, enhanced objectivity through quantitative methods (Lim et al., 2022a), reduced duplication, and support for researchers in refining their ideas (Brika et al., 2021). The study employs VOSviewer (van Eck and Waltman, 2010) and the Biblioshiny package in RStudio for bibliometric analysis (Aria and Cuccurullo, 2017). Finally, we assess the full-text content of the selected articles identified through the SLR to understand how the existing literature conceptualizes PT and its application in empirical research (Haggarty, 1996; Prashar and Sunder, 2020).

3.2. Search protocol

3.2.1. Search strategy

An SLR has a carefully defined set of criteria for article selection and involves multiple stages of quality assessment and relevance checks to identify and include all pertinent articles (Paul et al., 2021). Fig. 1 shows the methodology followed. We start by conducting the literature search using the Scopus database. Scopus is a comprehensive database of multidisciplinary journals and has been a reliable source extensively used in prior bibliometric and review studies (e.g., Bhukya et al., 2022; Brahmi et al., 2023, 2024; Ferreira, 2018). The rationale of limiting our search to the Scopus database is:

- i. Scopus is a repository of over 20,000 peer-reviewed journals from Elsevier, Springer, Emerald, Taylor & Francis, Wiley, INFORMS, and Inderscience (Gupta et al., 2021).
- ii. Scopus was selected over other databases, such as Web of Science (WOS, henceforth) and Google Scholar, for two main reasons: (a) Scopus ensures better representativeness compared to WOS, which indexes fewer journals due to its stringent criteria (Paul et al., 2021) and (b) While Google Scholar matches Scopus in terms of completeness, Scopus fares better than Google Scholar in excluding articles from predatory journals (Paul et al., 2021).

The search parameters included the article title, keywords, and abstract sections (Vrontis et al., 2020). A comprehensive search string was created by combining various keywords using Boolean operators (OR/AND). Therefore, the search string used was (Prospect theory OR Cumulative prospect theory OR Prospect theory model OR Cumulative prospect theory model). The study's search period is from 1979 (the base paper on PT was published in 1979) to 2024. The initial search yielded 5833 articles. A robustness check, executing the same search string on WOS for the 1979–2024 window, is reported in Appendix A. We see that the top contributing journals are broadly consistent across both databases, confirming that the Scopus-based corpus captures the same dominant publication outlets for PT research in finance, and that limiting our search to one database does not result in the loss of any significant studies.

⁵ Tranfield et al. (2003) note that, "SLR differs from traditional narrative reviews by adopting a replicable, scientific and transparent process, in other words, a detailed technology, that aims to minimize bias through exhaustive literature searches of published and unpublished studies and by providing an audit trail of the reviewers' decisions, procedures and conclusions."

3.2.2. Exclusion criteria

The initial collection of articles was further refined using a sequential set of exclusion criteria, ensuring only the most relevant and highest-quality publications were retained for analysis. Primary Screening: Following the principles of SLR, the 5833 records were subjected to three simultaneous metadata-based filters. First, the search was restricted to full English-language documents, thus ensuring broad accessibility and relevance (Baker et al., 2020; Lim et al., 2022b). Second, only journal articles were retained, excluding conference proceedings, book chapters, books, letters, and editorials. Third, records were restricted to the Scopus subject classification of Economics, Econometrics and Finance, ensuring domain relevance. The application of these three primary filters: language ($n = 386$ excluded), document type ($n = 1295$ excluded), and subject area ($n = 2891$ excluded), reduced the sample to 1261 articles.

Secondary Screening: Articles were further filtered using the Australian Business Deans Council (ABDC) 2022 Journal Quality List (<https://abdc.edu.au/abdc-journal-quality-list/>). Articles ranked ABDC A*, A, and B were considered for the analysis in this study, reducing the sample to 881 articles.

3.2.3. Selecting relevant articles

The pool of 881 articles was then assessed for relevance at the eligibility stage. The selection process involved reviewing the titles, abstracts, introductions, and conclusions, and the full paper when in doubt about relevance, to ensure PT was the core subject of each article. This process resulted in a seed list of 214 articles. An additional four articles were identified by manually screening the reference lists of included articles through backward and forward citation searches, following PRISMA 2020 guidelines (Page et al., 2021). At this stage, we scanned the corpus for any duplication and found none. Hence, 218 research studies have been considered for the bibliometric analysis.

Of these 218, we restrict our content analysis to ABDC A* and A, leaving a corpus of 126 articles.⁶ The rationale for this two-tier design is as follows. First, the broader bibliometric corpus ($n = 218$) increases the statistical robustness of the bibliometric mapping (co-citation networks, keyword co-occurrence, country and author maps). Second, content analysis requires close reading and critical synthesis at the level of the study. Such a level of rigor scales poorly with the number of articles. Moreover, restricting the content analysis to ABDC A* and A journals is not done arbitrarily. Journal quality in finance is a well-accepted proxy for methodological rigor, peer-review standards, and influence on subsequent research (Currie and Pandher, 2011; Engemann and Wall, 2009). The A*/A corpus, therefore, captures the papers that have most substantively shaped the theoretical and empirical development of PT in finance. This two-tier design is methodologically aligned with the SPAR-4-SLR protocol of Paul et al. (2021), which permits differential filtering across the descriptive (bibliometric) and analytical (content) stages of a hybrid review. Recent studies have adopted similar staged approaches, conducting bibliometric mapping on a broader corpus while restricting in-depth content analysis to high-quality (e.g., ABDC A*/A) articles (Jain et al., 2024; Janani et al., 2026).

As shown in Table 2, the study employs a selective approach with well-defined search criteria to refine and limit the final results. The qualitative literature review follows a structured three-step process: (1) conducting a literature search within a predefined database, (2) analyzing the literature and performing forward/backward searches to identify additional relevant publications, and (3) cataloging the resultant literature.

Following the outlined methodology, this study addresses five principal RQs: (a) What is the scope of research within the PT literature? (b) Which aspects of the literature, including documents, authors, sources,

⁶ As a robustness check, Appendix C presents the bibliometric analysis conducted on the A*/A corpus ($n = 126$). The findings broadly remain the same.

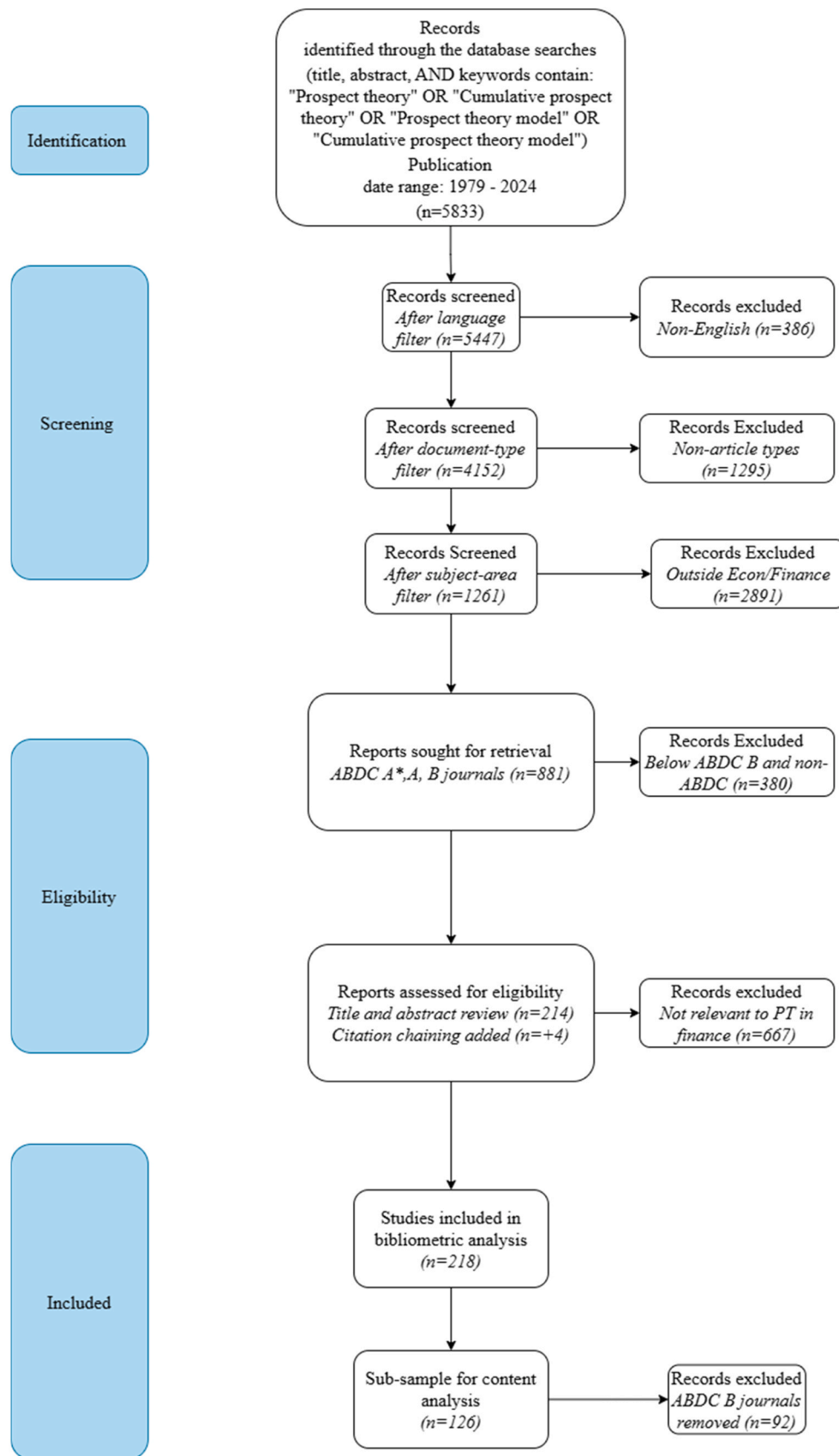


Fig. 1. Search strategy and the data retrieval process following Paul et al. (2021). **Notes:** Flow constructed following PRISMA 2020 (Page et al., 2021) and SPAR-4-SLR (Paul et al., 2021). Search period: 1979–2024.

Table 2

SLR Research protocol. The table documents the systematic review process following Briner and Denyer (2012).

Background and motivation • Growing evidence of investors' irrationality and departure from Expected Utility Theory predictions • Understand the role of prospect theory in the financial markets. • Need to consolidate and map the expanding PT literature in finance, given that 72% of the 218 papers in the prior review were published after the last comprehensive review (Barberis, 2013). Objectives • To know what can be learned from the existing literature about PT in the financial market. • To explore the applications of PT in understanding various financial market phenomena. Criteria for considering studies for this review • Studies that contribute to the understanding of PT in the financial market • Only peer-reviewed journal articles published or available online since the inception of PT (1979) were selected. • The subject area chosen was Economics, Econometrics, and Finance • Includes only journal articles Search strategy for identification of studies • Scopus database • Time period: 1979–2024 • Search terms and keywords- Prospect Theory OR Cumulative Prospect Theory OR Prospect Theory Model OR Cumulative Prospect Theory Model • Only English-language journal articles are considered. • Only published articles are considered. Eligibility • The inclusion criteria • Prospect Theory • Cumulative Prospect Theory • Loss aversion • Lottery demands • Market anomalies • The exclusion criteria • Insurance • Labour market • All articles that do not refer to PT or any component of PT Data collection • Scopus database Assessment of methodological quality • Australian Business Deans Council (ABDC) journal ranking list of 2022 Synthesis • Aggregation, synthesis, and interpretation

and countries, have been most influential? (c) What are the most frequently used terms in the field, and how do they cluster thematically? (d) What patterns can be observed in the citation network of the literature? (e) What future research directions does the current literature suggest?

4. Results from bibliometric analysis (RO 1 and RO 2)

Bibliometric findings are organized into two categories (Fig. 2). Descriptive analysis includes publication trends, relevant sources, authors, countries, and the most-cited articles (Donthu et al., 2021). (ii) Network analysis includes different statistical techniques applied to map the conceptual and intellectual structures of the bibliographic data.

To give the discussion section a cleaner structure, we organize the bibliometric results using a three-layered approach. Layer 1 (theoretical components) incorporates the PT's value function and PW function. Layer 2 (behavioral mechanisms) includes loss aversion, the disposition and endowment effects, and lottery-like preferences. Layer 3 (empirical applications) captures cross-sectional return prediction, the EPP, and asset-class-specific pricing anomalies.

4.1. Annual scientific production (RQ 1)

Fig. 3 illustrates the yearly output of peer-reviewed PT articles from its inception until 2024. The increasing number of published articles highlights researchers' growing interest in this field. Although the PT originated in 1979 with the seminal work of Kahneman and Tversky (1979), significant growth in the literature began in the late 1990s and early 2000s, coinciding with the mainstream acceptance of behavioral finance (Benartzi and Thaler, 1995; Shiller, 2003). One important observation here is that this growth pattern in PT literature has not been uniform. Since the 2010s, the literature has focused on asset-pricing

applications of PT, while the growth of foundational behavioral themes, such as the EPP, has been relatively sluggish. In other words, while certain sub-themes have reached maturity, some emerging areas remain. This dichotomy has implications for the future research agenda (Section 7).

4.2. Most relevant sources (RQ 2)

Table 3 shows that the key journals driving research in PT are from both the decision science and finance domains. This cross-disciplinary distribution tends to introduce heterogeneity in methodological standards and empirical rigor across the corpus. Core finance journals such as the "Journal of Banking and Finance" and the "Journal of Financial Economics" feature prominently, indicating PT's integration into the mainstream finance literature and underscoring the need to synthesize the literature. Furthermore, the top 14 journals have published 95 articles, accounting for about 43% of the corpus.

4.3. Most cited articles (RQ 2)

Table 4 presents the most prominent studies in the literature, ranked by global citations, highlighting their impact. "Prospect Theory: An Analysis of Decision under Risk", which introduced PT, leads with 83,643 citations; followed by "The Framing of Decisions and the Psychology of Choice" and "Advances in Prospect Theory: Cumulative Representation of Uncertainty". The highest citations to Tversky and Kahneman's work signal that, while PT's theoretical core is well acknowledged, the empirical literature applying PT to financial markets remains more fragmented. Conflicting findings on, for instance, the magnitude of the loss aversion coefficient across demographic groups and market contexts (Abdellaoui et al., 2008; Booij et al., 2010) are not addressed by the high-citation foundational papers.

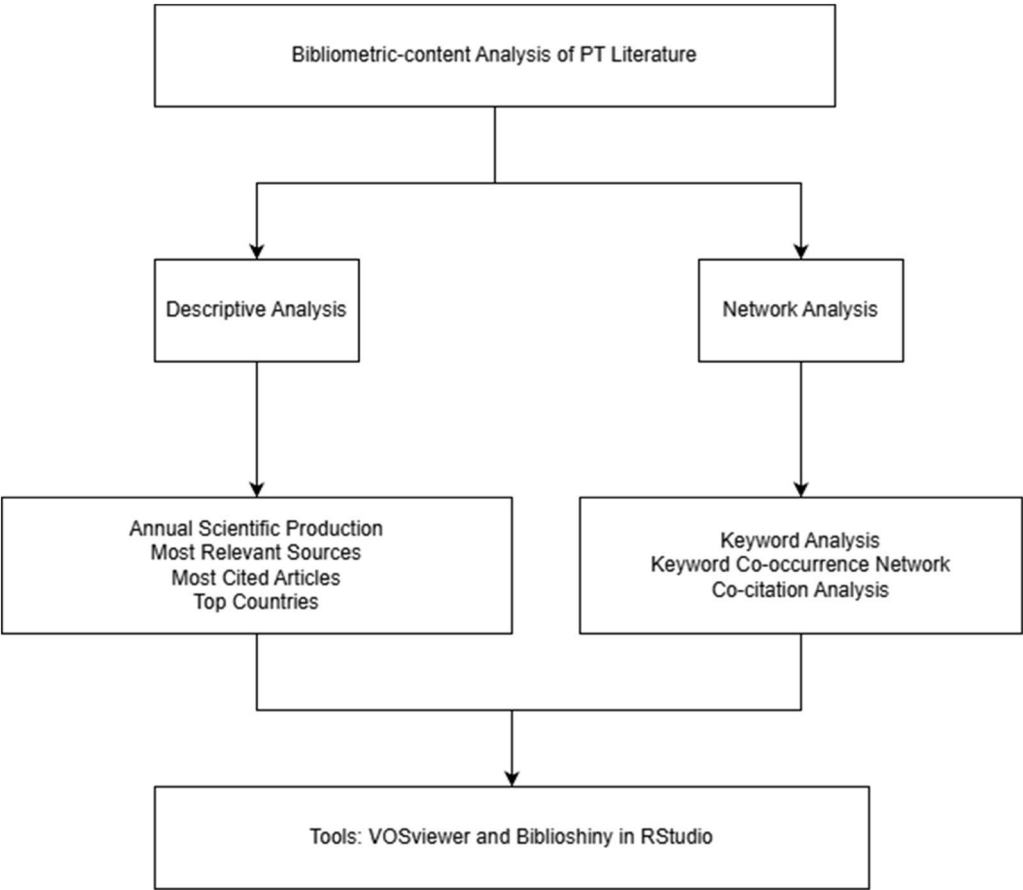


Fig. 2. Analysis structure of bibliometric results (created at "https://app.diagrams.net/").

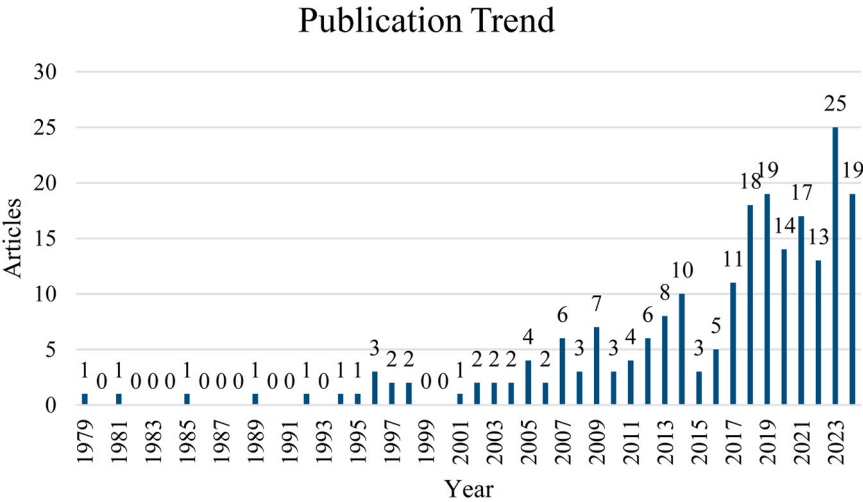


Fig. 3. Annual Scientific Production. **Notes:** Source of data = Scopus; Period of coverage = 1979–2024; Total publications = 218. Source: Result from the Biblioshiny app (app for Bibliometrix in RStudio); Author's own creation.

4.4. Authors' map and most influential authors/ (RQ 2)

Fig. 4 shows the extent of collaboration among the influential authors in the field. The inclusion criteria are a minimum of 2 publications per author. The network comprises several distinct, internally connected clusters with no visible inter-cluster edges. This suggests that PT research in finance has developed through parallel, siloed efforts rather than through collaborations amongst the scholars. The limited presence

of international collaborative clusters also reinforces the geographic concentration observed in section 4.5. Table B1 ranks the top 10 authors by publications and citations.

4.5. Leading countries for research on PT (RQ 2)

Fig. 5 presents the country-level co-authorship network. A minimum threshold of 2 publications per country was applied for inclusion in the

Table 3

Most relevant sources.

Rank	Journal	Publisher	Publications	Citations	ABS rank	Scimago	ABDC rank
1	Journal of Risk and Uncertainty	Springer International Publishing	9	11,267	3	Q1	A*
2	Journal of Financial Economics	Elsevier	5	1146	4*	Q2	A*
3	Journal of Economic Behavior and Organization	Elsevier	11	744	3	Q1	A*
4	Theory and Decision	Springer International Publishing	9	384	2	Q1	A
5	Journal of Banking and Finance	Elsevier	6	195	3	Q1	A*
6	Journal of Behavioral Finance	Taylor & Francis Online	8	152	2	Q3	A
7	Review of Behavioral Finance	Emerald Group Publishing	8	84	1	Q3	B
8	Journal of Behavioral and Experimental Finance	Elsevier	5	67	1	Q1	A
9	Computational Economics	Springer International Publishing	6	64	2	Q2	B
10	Pacific-Basin Finance Journal	Elsevier	7	61	2	Q1	A
11	Managerial Finance	Emerald Group Publishing	6	45	1	Q3	B
12	Investment Management and Financial Innovations	Business Perspectives Ltd	5	32	N/L	Q3	B
13	Risks	MDPI	5	22	N/L	Q2	B
14	Applied Economics Letters	Taylor & Francis Online	5	8	1	Q3	B

Notes: ABDC ranking is as per the Australian Business Deans Council 2022 Journal Ranking List. ABS rank is as per the Association of Business Schools 2021 Journal Quality Guide. Scimago is SJR: Scientific Journal Rankings (2024)

Source: Result from VOSviewer app; Author's own creation

Table 4

Most cited articles.

Rank	Study	Title	Source Title	Cites	DOI
1	Kahneman and Tversky (1979)	Prospect theory: An analysis of decision under risk	<i>Econometrica</i>	83643	10.2307/1914185
2	Tversky and Kahneman (1981)	The Framing of Decisions and the Psychology of Choice	<i>Science</i>	12414	10.1126/science.7455683
3	Tversky and Kahneman (1992)	Advances in Prospect Theory: Cumulative Representation of Uncertainty	<i>Journal of Risk and Uncertainty</i>	10941	10.1007/BF00122574
4	Odean (1998)	Are investors reluctant to realize their losses?	<i>Journal of Finance</i>	1957	10.1111/0022-1082.00072
5	Shefrin and Statman (1985)	The Disposition to Sell Winners Too Early and Ride Losers Too Long: Theory and Evidence	<i>The Journal of Finance</i>	1842	10.1111/j.1540-6261.1985.tb05002.x
6	Benartzi and Thaler (1995)	Myopic loss aversion and the equity premium puzzle	<i>Quarterly Journal of Economics</i>	1452	10.2307/2118511
7	Barberis et al. (2001)	Prospect theory and asset prices	<i>Quarterly Journal of Economics</i>	812	10.1162/003355301556310
8	Barberis and Huang (2008)	Stocks as lotteries: The implications of probability weighting for security prices	<i>American Economic Review</i>	848	10.1257/aer.98.5.2066
9	Loughran and Ritter (2002)	Why Don't Issuers Get Upset About Leaving Money on the Table in IPOs?	<i>Review of Financial Studies</i>	787	10.1093/rfs/15.2.413
10	Grinblatt and Han (2005)	Prospect theory, mental accounting, and momentum	<i>Journal of Financial Economics</i>	548	https://doi.org/10.1016/j.jfineco.2004.10.006

Source: Result from VOSviewer app; Author's own creation

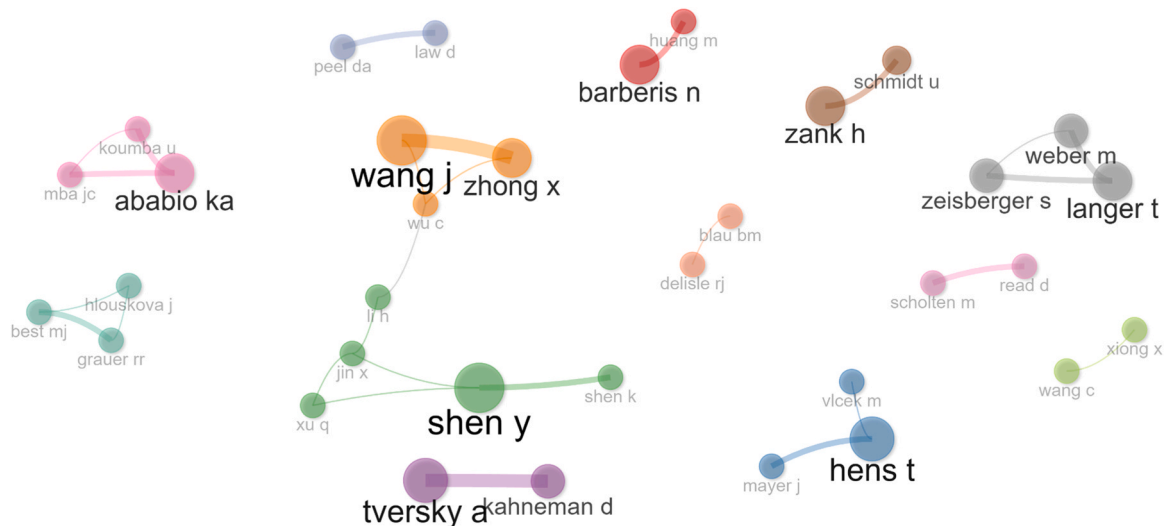


Fig. 4. Authors' collaboration network. Source: Result from the VOSviewer app.

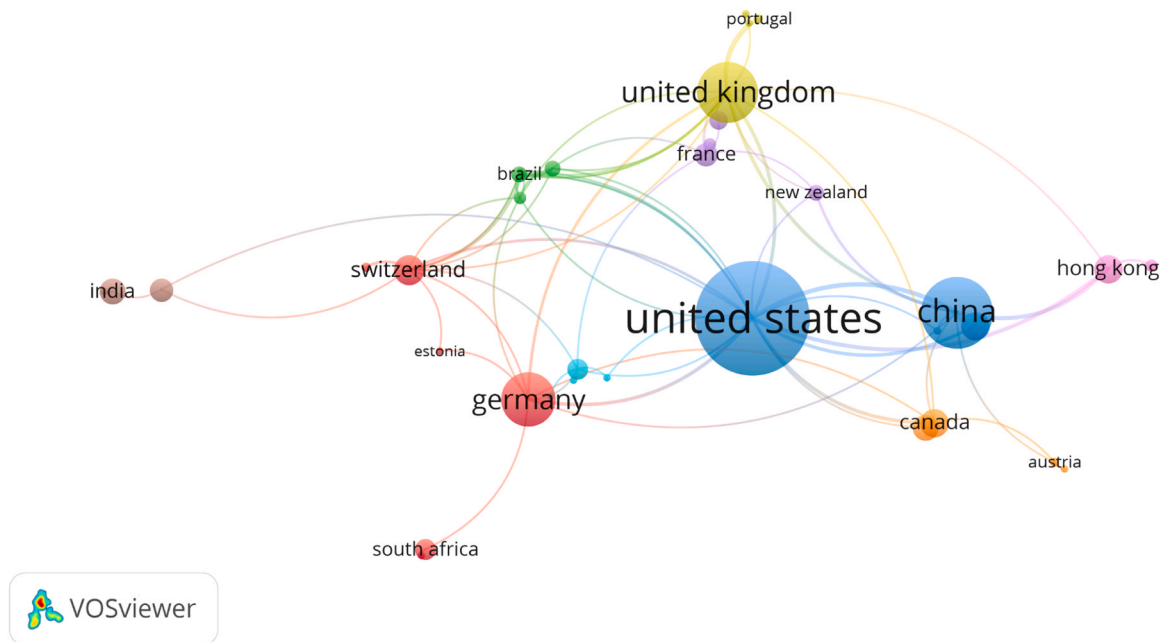


Fig. 5. Countries map. Source: Result from the VOSviewer app.

map. The node size reflects publication volume, and the edge weight reflects collaborative linkages between countries. We observe a concentration of research output in three countries (the US, China, and the United Kingdom). Germany emerges as a secondary hub, followed by others, including Switzerland, France, Canada, and Hong Kong. Among emerging economies, India, Brazil, South Africa, and Estonia feature; however, the nodes show limited connectivity to the core network, highlighting a gap for future primary research. Table B2 provides dual-panel rankings by total citations (Panel A) and total publications (Panel B).

4.6. Keyword analysis (RQ 3)

Fig. 6 displays the word cloud with prevalent keywords, highlighting

key themes in PT research. The words like "prospect theory", "cumulative prospect theory", "loss aversion", "probability weighting", and "disposition effect" appear more frequently than the rest. This indicates that most work on PT in finance focuses on its individual components. The relative infrequency of terms such as "gender differences" and "lottery stocks" identifies empirically underexplored themes within the PT-finance literature.

4.7. Keyword co-occurrence analysis (RQ 3)

In Fig. 7, each node size corresponds to word frequency in the corpus, with larger nodes indicating greater prominence in the literature. Each color represents a different theme. Words appearing together in the same article title are linked, with thicker, darker connections indicating

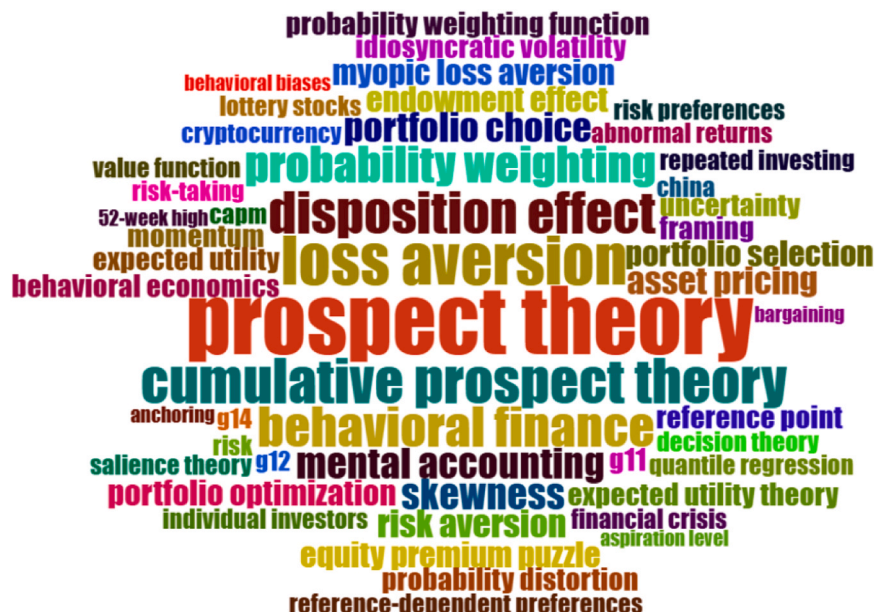


Fig. 6. Keyword analysis. Source: Result from the Biblioshiny app (app for Bibliometrix in RStudio).

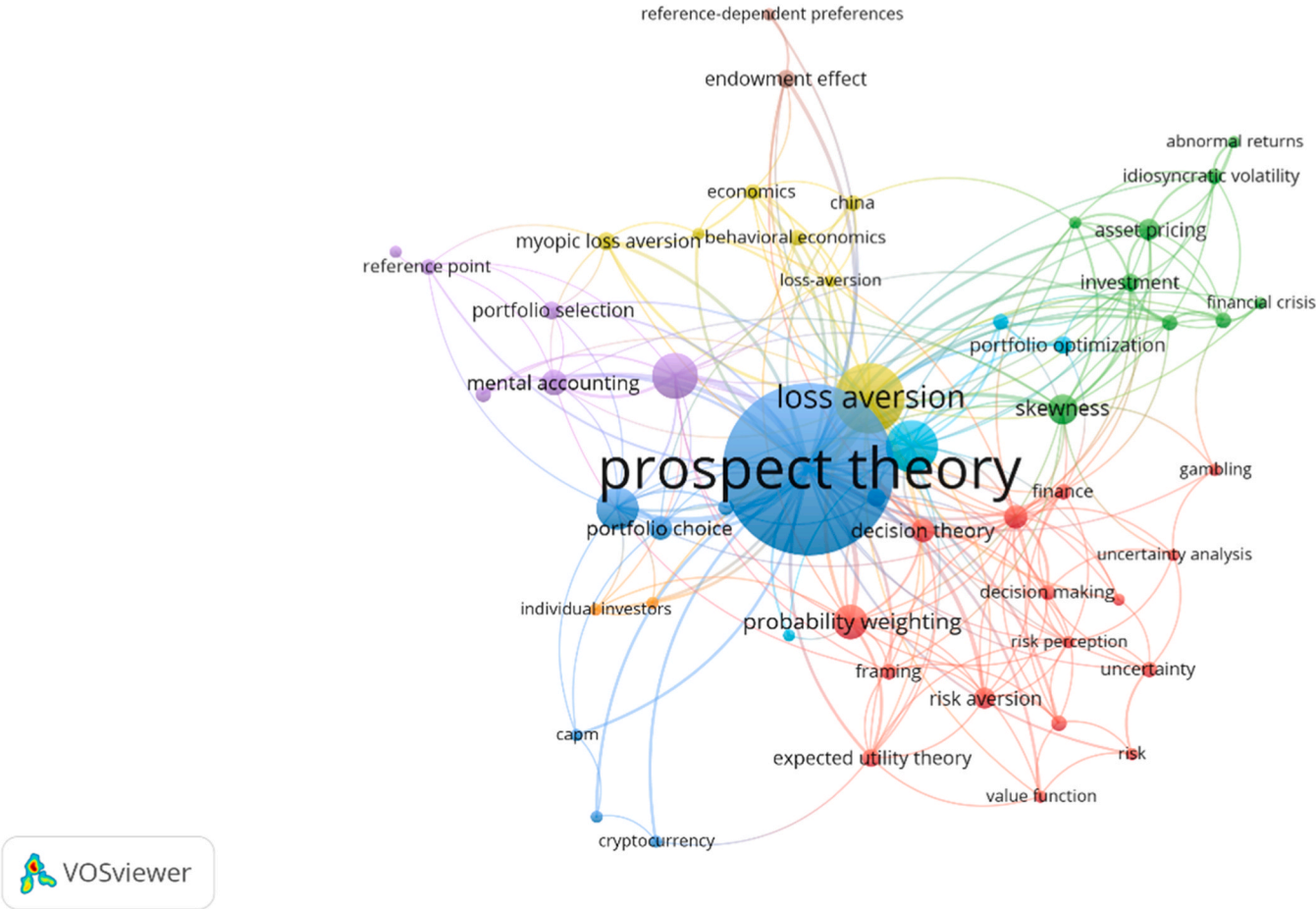


Fig. 7. Keyword co-occurrence analysis. Source: Result from the VOSviewer app.

more frequent co-occurrence. Key thematic clusters are identified:

- Blue: "Prospect theory" is the largest node beside "decision theory" and "portfolio choice".

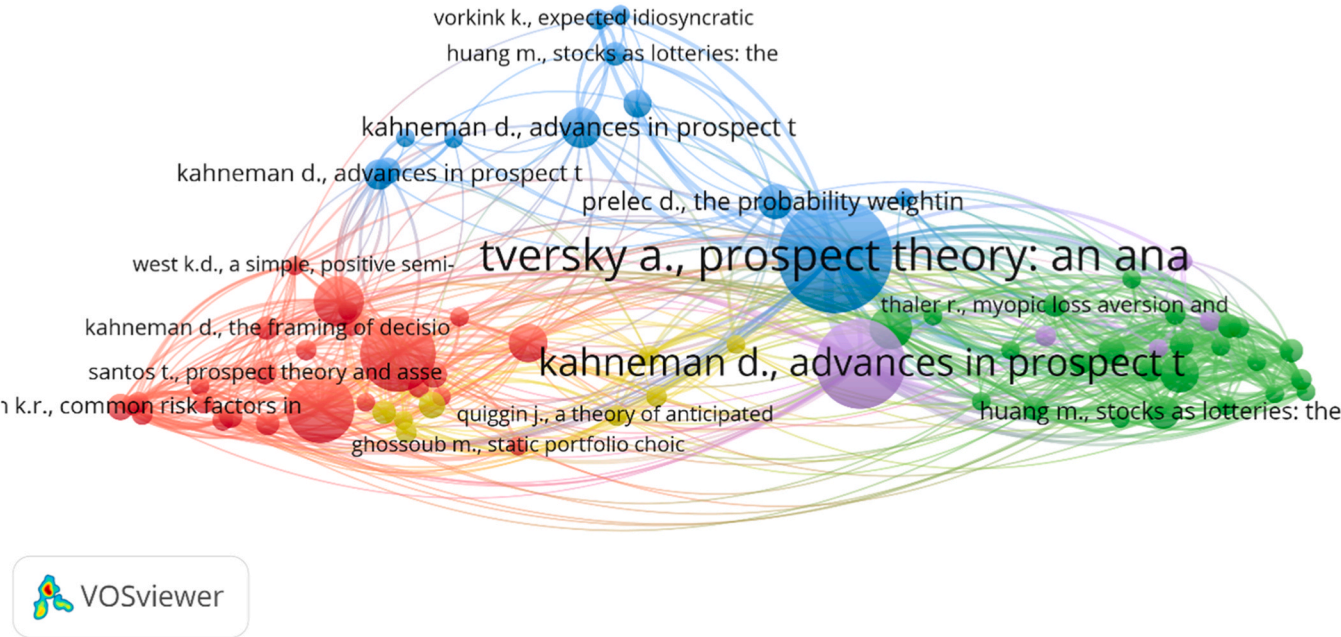


Fig. 8. Co-citation analysis. Source: Result from the VOSviewer app.

- Yellow: Relates "loss aversion" with "myopic loss aversion," "behavioral economics," "endowment effect," and "reference-dependent preferences".
- Green: Groups "asset pricing," "portfolio optimization," "skewness," "idiosyncratic volatility," and "abnormal returns".
- Red: Centered on "probability weighting," this cluster includes "Expected utility theory," "value function," "framing," "risk aversion," and "uncertainty".
- Purple: Groups "mental accounting," "reference point," "portfolio selection," and "individual investors".
- Light Blue: Contains "CAPM" and "CC".

The six clusters make a three-layer structure. Blue and Red jointly form Layer 1. Yellow and Purple form Layer 2 and are conceptually inseparable; Green and Light Blue form Layer 3. Skewness and idiosyncratic volatility are priced through probability weighting (PW, henceforth), while return anomalies are jointly priced through loss aversion and PW (Barberis et al., 2016; Barberis and Huang, 2008). The yellow cluster's central position reflects its role as the primary bridge for transmitting PT components into asset-pricing applications.

4.8. Co-citation analysis (RQ 4)

Co-citation analysis helps identify the key papers that influence the PT literature. The key nodes in Fig. 8 correspond to the two seminal papers: "Advances in Prospect Theory: Cumulative Representation of Uncertainty" and "Prospect Theory: An Analysis of Decision under Risk." It also shows five distinct clusters, each represented by a different color, of the field's most frequently cited papers. The five co-citation clusters reveal that the empirical literature on PT is quite dispersed, with distinct clusters forming around asset pricing applications, behavioral investor studies, laboratory-based utility estimation, and market anomalies research. This structural dispersion reflects the productive but fragmented nature of PT's empirical literature in finance, and reinforces the value of the present synthesis in establishing cross-cluster connections.

5. Content analysis (RQ 1; RQ 5 and RO 2)

This section expands on the key themes identified in the bibliometric analysis. We start with the application of PT in financial markets, focusing on its contributions to asset pricing and the cross-section of returns. Next, we then turn to specific components of PT that account for important investor behaviors and enduring market puzzles, including preferences for lottery-like assets, the EPP, the disposition effect, and the endowment effect.

5.1. Cross-section of returns

An imperative question recently attracting investors' attention is whether the PT can explain the cross-sectional variations in expected returns. Because opportunities to observe relevant probability distributions outside of laboratory settings are rare, the broader application of the PT preference functional has been limited. As a result, for a long time, most empirical studies focused on specific aspects of PT, such as loss aversion (Kahneman et al., 1991; Haigh and List, 2005), reference dependence (He and Zhou, 2014; Koszegi and Rabin, 2007), and PW (Barberis and Huang, 2008; Fehr-Duda et al., 2006). Later, Barberis et al. (2016) formulated a two-step model of noise trader behavior and derived a measure of PTV based on a stock's historical return distribution. They employed PTV as a cross-sectional predictor of returns, and subsequent studies used it. This makes it possible to compare the findings of different studies, but it also carries forward their limitations. Further, the literature also indicates that investors across countries exhibit different psychological biases influenced by cultural values (for example, Chui et al. 2010 and Kumar 2009). This heterogeneity is a persistent theme in the market and asset-specific evidence field that

follows.

5.1.1. Equity

In the United States (US, henceforth) equity market, Barberis et al. (2016) demonstrate that stocks with higher PTVs are preferred by investors, leading to their overvaluation and, hence, lower subsequent returns. Consistent with the limits-to-arbitrage hypothesis, the relationship is more robust in small, illiquid, and volatile stocks. Wang et al. (2021) note that the return-PTV relationship becomes significant only in the Chinese B-share market after it opens to domestic individual investors, suggesting that the PTV effect is stronger in markets where individual investors dominate. Though these studies offer consistent findings, the picture becomes somewhat blurred when tested in other countries, thus requiring a more critical review of the literature

For example, in collectivist societies such as South Korea, investors hold irrational beliefs about the upside potential of stocks with lower PTVs (Son and Nguyen, 2019). The findings contrast with the negative return-PTV relationship reported by Barberis et al. (2016) in the US market, which ranks higher on the individualism score. Furthermore, in Thailand, where the investor base is highly religious, PTV predicts future returns with a negative correlation (Chen et al., 2022c).

Emerging markets show varied results: Russia, China, and Brazil exhibit a negative return-PTV relationship, while Mexico shows a positive correlation, and in South Africa, results are mixed (do Nascimento Junior et al., 2021). In a similar vein, Eom and Park (2020) document that the return-PTV relationship also varies across different stock return distributions; for instance, stocks with fat-tailed distributions attract more attention from investors with PT preferences. In the Chinese market, the negative relationship between PTV and returns is pronounced in stocks far from their 52-week highs (Chen et al., 2022a), underscoring the relevance of the anchoring effect.

5.1.2. Mutual Funds

The dominance of retail investors in the mutual fund market (Wang and Han, 2023) makes it a natural setting for testing PT preferences and to examine whether PTV-driven investor behavior distorts fund flows and performance, and through which channels. In this regard, several studies document that funds with high PTV tend to attract higher net inflows and offer lower subsequent returns (Guo and Schoenleber, 2020; Han et al., 2026). This pattern is consistent with PT-driven overvaluation and is labeled the "dumb money effect" (Gu and Yoo, 2021; Gupta et al., 2022; Wang and Han, 2023). Further, the relationship strengthens during periods of high investor sentiment (Gu and Yoo, 2021). The relationship persists even after accounting for prominent behavioral explanations such as return extrapolation and salience theory (Han et al., 2026), underscoring the novel information that PT conveys.

In contrast, Yu et al. (2022) suggest that despite the PT preferences, high PTV funds do not necessarily become overvalued and continue to offer higher net inflows in the future. They attribute this to the lack of an agreed-upon intrinsic value for a fund. Another potential explanation is that the lack of efficient arbitrage opportunities makes it difficult to correct this mispricing. This conflict remains largely unresolved to date. Complementing this mixed evidence, Jin et al. (2023) document that the disposition effect among Chinese mutual-fund investors weakened significantly during the COVID-19 pandemic. They attribute this decline to heightened risk aversion, liquidity-driven forced sales, and reduced self-image concerns when peer performance is relatively strong.

5.1.3. Bonds

Historically, Benartzi and Thaler (1995) propose that investors consider a corporate bond's historical returns in the way that PT suggests. Zhong and Wang (2018) find results similar to those of Barberis et al. (2016) in the equity market, but they further conclude that, unlike the stock market, the loss aversion component primarily drives PT's predictive power in the bond market. This aligns with existing literature suggesting that institutional investors are more loss-averse than retail

investors, as they care more about losing the respect amongst their peers (Barberis and Huang, 2001; O'Connell and Teo, 2009).

All in all, the bond market evidence rests on a very narrow empirical base (Benartzi and Thaler, 1995; Zhong and Wang, 2018). Zhong and Wang (2018) is restricted to the US investment-grade and junk bond universe, and their claim that loss aversion drives PT's predictive power in bonds has not been replicated in non-U.S. bond markets, nor in sovereign bond markets, where the investor base and credit-risk structure differ from those of corporate bonds. Furthermore, the institutional investor and loss-aversion argument invoked to explain why bonds differ from equities is, though supported by theory and limited experimental evidence, untested with bond-specific trading data.

5.1.4. Cryptocurrencies

CCs do not have an underlying fundamental value; thus, how individuals make investment decisions significantly impacts the CC returns. Chen et al. (2022b) tested the predictions of PT in a large sample of 1573 CCs over the period 2014–2020, and their findings corroborate PT's predictions in conventional markets such as stocks (Barberis et al., 2016) and bonds (Zhong and Wang, 2018). They further conclude that PTV predictions are more robust among difficult-to-arbitrage CCs, which are smaller, highly volatile, and illiquid. Similar findings are reported by Thoma (2021) and Yadav (2024). However, Chen et al. (2022b) sample period spans several structural changes in the market, including the 2017–2018 bubble and the COVID–19 shock, which are not controlled for as structural breaks. The PTV measure is applied directly from Barberis et al. (2016), inherits assumptions about return distributions (e. g., stationarity, continuous trading) that are less reliable for CCs than for equities, given the extreme non-stationarity and 24/7 trading structure of CC markets. Additionally, extant studies argue that the retail-dominated, sentiment-driven nature of the CC market makes it an ideal PT test bed. Still, this same feature makes it difficult to empirically disentangle the PT and sentiment signals.

5.2. PW: lottery preferences among investors

The PW function predicts that investors overweight small probability, high-payoff events, creating excess demand for positively skewed assets, overpricing, and lower subsequent returns (Barberis and Huang, 2008). Empirical work across equities, over-the-counter stocks, options, and commodities aligns with this proposition (Ang et al., 2006; Bali et al., 2011; Eraker and Ready, 2015; Félix et al., 2019; Yuan et al., 2023). Now, the question is how robustly it can be attributed specifically to PW, and how stable it is across investor types, cultures, and market conditions. The impact of investor sentiment (Blau, 2017) and market downturns (Gong et al., 2021) on lottery premium is though consistent with PT, it is equally predicted by salience theory and other attention-based models. However, there has been no formal test to identify the true effect of these competing mechanisms. Moreover, there is an inconsistency in how the studies capture lottery preference. For instance, ex-ante skewness (Conrad et al., 2013), MAX effect (Bali et al., 2011), idiosyncratic volatility (Ang et al., 2006), which are not equivalent representations of the PW mechanism in PT, and the choice of measure materially affects which stocks are classified as lottery-like and which market segments drive the results.

Furthermore, findings that women display lower sensitivity to probability changes than men (Fehr-Duda et al., 2006); collectivist cultures demand higher probability-distortion premia (Hollstein and Sejdiu, 2023); seasonal mood amplifies lottery-stock anomalies, though internally coherent, have several limitations. For example, the cultural operationalizations differ across studies, the gender finding rests on laboratory data, and no unified framework explains when and why PW heterogeneity produces economically meaningful return differences versus noise. Henderson et al. (2018) provide a dynamic optimal-stopping model and show that PW, along with the value function, generates both widespread use of stop-loss thresholds on losses and

a long right tail on gains-behavior. This behavior rationalizes observed stop-loss strategies and produces a disposition effect whose magnitude closely matches Odean's (1998) empirical benchmark.

In conclusion, though the literature establishes the existence of a lottery premium, it does not establish its mechanism.

5.3. Loss aversion

Loss aversion is defined as the value function being steeper for losses than gains (Kahneman and Tversky, 1979). When individuals assess the gain-loss value, loss aversion plays a central role. For example, the expected gain-loss value of a risky investment with a 50–50 chance of winning \$1000 or losing \$500 could be negative to an individual with a loss aversion of 4.0 ($1/2 \times \$1000 - 4 \times 1/2 \times \$500 = -\$500$), but positive to an individual with a loss aversion of 1.5 ($1/2 \times \$1000 - 1.5 \times 1/2 \times \$500 = +\$125$). Therefore, an individual with a high loss aversion is more likely to decline the investment opportunity and maintain the status quo. Loss aversion is characterized by individuals being more sensitive to decreases in their well-being than increases. Further, the value function in PT has a kink at the origin, which makes the loss function steeper than the gain function. The ratio of these slopes captures the degree of loss aversion.

5.3.1. Loss aversion and EPP

Plausible levels of risk aversion under EUT fail to explain the historically large and continual gap between equity returns and risk-free rates (Mehra and Prescott, 1985). Myopic loss aversion, which they define as the combination of loss aversion and frequent portfolio evaluation, has been proposed to explain this effect (Benartzi and Thaler, 1995). Later, Barberis et al. (2001) formalized a dynamic asset-pricing model to explain this equity premium.

Over the years, the research has refined and qualified these early claims. Barberis (2013) concluded that, though loss aversion helps explain the EPP, myopic loss aversion alone is insufficient. Li and Yang (2013) incorporated full PT preferences (loss aversion and PW) into a general-equilibrium framework to reconcile these equity premium magnitudes. In parallel, Ert and Erev (2013) provide six clarifications on the descriptive validity of loss aversion in decisions under risk. They demonstrate that the coefficient commonly used in asset-pricing testing ($\lambda = 2.25$ from Tversky and Kahneman 1992) is highly context dependent. Later, Blavatsky (2021) shows that measured loss aversion is highly sensitive to methodological choices. With these advancements in the literature, the debate has shifted to "Is the loss-aversion parameter sufficiently well-defined and stable to explain observed EPP?" rather than "Does loss aversion explain the EPP?" More recently, Morais et al. (2023) show that the aggregation effect on myopic loss aversion exhibits diminishing returns and that participants learn over repeated trials. They suggest that as investors gain experience with the underlying lottery, the incidence of myopic loss aversion declines.

Conclusively, from the above discussion, the following key observations emerge. First, the explanatory power of loss aversion for the EPP is not static, and PW plays an important role in explaining the magnitudes of the premia (Li and Yang, 2013). Second, the literature is increasingly arguing that λ is context-dependent, which weakens the fixed-parameter assumptions of earlier models (Blavatsky, 2023; Ert and Erev, 2013). Third, the literature has yet to converge on a single parameter value of λ that explains the historical equity premium. These unanswered questions highlight both the opportunities and the current limitations of PT as an explanation for the EPP.

5.3.2. Disposition effect

Besides the EPP, the most concrete and well-documented evidence of loss aversion in financial markets is the tendency of investors to hold losing investments. The phenomenon of "sell winners too early and hold on to losers too long" (Shefrin and Statman, 1985) has been observed across retail investors (Odean, 1998), professional traders, and markets.

The classical explanation combines PT's S-shaped value function with mental accounting (Kahneman and Tversky, 1979; Thaler, 1980). Post-2013 research has focused on whether PT still creates the disposition effect once fully specified and whether the effect is static. Li and Yang (2013) conclude that diminishing sensitivity predicts a disposition effect, price momentum, a reduced return volatility, and a positive return-volume correlation. Loss aversion, by contrast, generally predicts the opposite. In calibrated economies, there is a significant range of preference parameters for PT to simultaneously explain the disposition effect, the momentum effect, and the EPP. This implies that the popular claim that "PT explains the disposition effect" is misleading, and the outcome depends on which PT component dominates. Later on, Talpsepp et al. (2014) combined laboratory experiments with real stock-market trading data. They find that domestic individual investors exhibit the strongest disposition effect, while foreign institutional investors show an inverse disposition effect. They note that this pattern cannot be explained by PT alone. Plotnicki and Szyska (2014) in the Initial Public Offering (IPO, henceforth) market document an analogy between firms (debuting too fast in a bullish market and too slow in a bearish market) and investors, and conclude that the disposition effect holds for corporate managers as well as investors in the IPO market. Furthermore, Eom (2018) analyses intraday data from the Korean stock index futures market and identifies an "opposite disposition effect". They state that the disposition effect depends on the size of unrealized gains and losses, and that it applies only below a certain threshold: investors hold losers longer than winners. Wang et al. (2023) investigate the disposition effect among managers regarding the stocks of their own companies when they are insiders. They state that the managerial disposition effect cannot be entirely explained by insider-selling constraints and reflects managers' willingness and preference for realizing more paper gains. The superior information held by managers only partially corrects the disposition effect.

In conclusion, the following important observations emerge from the post-2013 literature. First, PT predicts the disposition effect only under restrictive parameter configurations (Li and Yang, 2013). Second, the direction of the effect is not universal and symmetric. Studies observe that inverse disposition effects appear among institutional investors (Talpsepp et al., 2014) and in futures markets above certain thresholds (Eom, 2018). Third, the bias extends beyond retail investors to corporate managers (Wang et al., 2023), weakening the hypothesis of irrationality among retail investors.

5.4. Endowment effect

Beyond the experimental setting (Knetsch and Sinden, 1984), the literature has more recently tied the concept to two components of PT: reference dependence (Smith, 2019) and loss aversion to explain the endowment effect. A related question in literature is whether reference points are simply lagged expectations (Koszegi and Rabin, 2007). Heffetz (2021) proposes that beliefs must "sink in" before they affect behavior. They argue that experimental tests that manipulate only lagged beliefs without ensuring sink-in might be understating the theory's explanatory power. Contrary to the classic belief, an individual's reference point is not fixed and is not necessarily determined by present circumstances. It can also be shaped by expected future outcomes and the history of the ownership (Strahilevitz and Loewenstein, 1998). People who expect to own an item update their reference point accordingly and frame the potential changes as losses. Conversely, those who do not expect to own the item use non-ownership as their reference point, framing potential changes as gains. This theoretical extension helps explain variations in the willingness-to-pay (WTP)-willingness-to-accept (WTA) gap (Birnbaum, 2018) and shows that expectations about future ownership can moderate these discrepancies.

The literature presents several important observations. First, the classical demonstrations are laboratory experiments that use low-stakes goods (mugs, chocolate bars) with student samples. Pryor et al. (2018)

note that though loss aversion influences riskless choice, translating mug-exchange experiments into financial asset valuation requires several assumptions. Second, the WTP-WTA gap is sensitive to experimental design choices, including whether subjects receive compensation, the order of elicitation, and whether subjects are experienced traders (List, 2003). Third, the theoretical extension proposed by Strahilevitz and Loewenstein (1998) generalizes the static loss-aversion explanation of the endowment effect by modeling the reference point as gradually shifting through adaptation to ownership history. However, this dynamic element remains unincorporated into financial market models. These limitations suggest that, while the endowment effect is well-documented experimentally, its precise mechanism and financial market implications remain largely underexplored. Araujo et al. (2022) prove that the endowment effect prevails only under low aggregate risk held by risk-averse agents. They show that when aggregate uncertainty is sufficiently large, PT agents leave the reference point and trade to a frontier allocation, hence overturning the classic endowment-effect prediction.

6. Limitations, insights, and future research agenda (RQ 5; RO 3 and RO 4)

6.1. Limitations

This section highlights the study's limitations and may further motivate the future research agenda. First, the literature search is based on the Scopus database and restricted to ABDC A*/A-ranked journals for content analysis. Doing so may underrepresent work published in regional and lower-ranked decision science journals (as ranked by the ABDC). Second, the search string is anchored on "prospect theory" and its variants; studies that apply PT components (e.g., loss aversion, PW) without explicitly naming the theory are not captured, even though they may offer important insights. Third, bibliometric techniques quantify patterns in the literature but cannot assess methodological rigor at the study level; the conflicting empirical findings identified across clusters (e.g., on the λ parameter) indicate where study-level meta-analysis remains necessary.

6.2. Insights and future research agenda

Charting pathways for future research is an important agenda for systematic reviews (Lim et al., 2022a). With the ever-increasing empirical examination of PT in the financial market, behavioral finance, which is now firmly established as a mainstream area within the broader finance literature, continues to benefit from advances in understanding investor decision-making. The content analysis in Section 5 uncovers a set of unresolved questions that define the agenda for future research. In this article, we identify several pertinent pathways from the contradictions documented in the content analysis and encourage future researchers to build on the findings of past research on PT that we have reviewed.

- **High-Frequency Data Analysis:** The extant literature on the role of PT in the cross-section of returns has primarily used low-frequency data (monthly, weekly, and daily). Modern financial markets, with their limit-order book microstructure and electronic communication networks, are extremely fast and capable of handling large volumes of trades. Future research should leverage this higher-granularity data, which offers several advantages. Moreover, the rise of High-Frequency Trading (HFT) has led to quick corrections of market anomalies within short time horizons (O'Hara, 2015). Thus, examining potential mispricing at the highest available frequency to capture and analyze these rapid market adjustments effectively is a potent avenue.
- **Parameter estimation and nonlinear modeling:** First, asset-pricing studies may attempt to capture and report sensitivity bands of

equity-premium and PTV-return predictions, computed across the empirical λ distribution documented in Ert and Erev (2013) and Blavatskyy (2021). Second, existing studies on PT predominantly rely on conditional mean models. However, as stock returns data often exhibit nonlinearities (high skewness and excess kurtosis), future studies should explore potential asymmetric relationships between PT and asset returns. In this regard, the quantile regression methodology (Koenker and Bassett, 1978, 1982) is attracting much attention from academicians.

- **Cultural and Demographic Influences:** Cross-country evidence reviewed in 5.2 indicates that PT predictions vary across cultural settings. However, the literature has yet to identify which cultural element primarily drives this variation. Thus, future research may explore the interactions between various cultural proxies and PT, examining how value systems shape PT preferences. Also, cultural characteristics play a significant role in shaping variations in how probabilities are perceived (Chui et al., 2010). Therefore, joint estimation of PT parameters across countries, projected onto cultural dimensions, would move the literature beyond single-proxy tests.
- **Reference Point Definition:** Though a foundational element of PT, empirical studies have, to date, often adopted an inappropriate specification. Doing so significantly impedes the practical application of PT in finance and is widely regarded as a major weakness in PT's theoretical framework (Fudenberg, 2006; Pesendorfer, 2006). Furthermore, a large body of research shows that cultural factors significantly impact individual judgment and decision-making. Thus, while aiming to establish a more precise definition of reference points, future studies should also investigate whether reference points exhibit cross-cultural variation and, if so, the possible causes of these variations.
- **Disposition Effect Reconciliation:** While loss aversion is commonly cited to explain the disposition effect, debates persist about the degree of convexity in the value function. Some research suggests a need for greater convexity than experimental evidence indicates (Li and Yang, 2013). Another explanation comes from "realization utility." Barberis and Xiong (2009) demonstrate that even a linear realization utility can produce a strong disposition effect if the time discount rate is sufficiently positive. Future studies should aim to harmonize these diverse viewpoints and consider additional behavioral factors to provide a comprehensive explanation of the disposition effect and observed trading patterns.
- **Reconciling the mutual fund overvaluation conflict:** The opposing findings on whether high-PTV funds become overvalued can be empirically tested. Joint estimation of the PTV-flow and PTV-performance relationships, conditional on fund-level proxies for arbitrage frictions and the share of institutional ownership, will help in identifying the moderating effects.
- **Gender Disparities:** While a growing body of research has examined gender-based differences in behavioral biases under PT (e.g., Fehr-Duda et al., 2006; Kean-Siang et al., 2021; Ladrón de Guevara Cortés et al., 2023; Sarin and Wieland, 2016), the literature remains fragmented and context-dependent. Existing evidence suggests that gender differences in risk preferences vary with the source of uncertainty, the magnitude of stakes, and the cultural setting in which decisions are made. Future research may extend this work by (i) jointly estimating all PT parameters ($\alpha, \lambda, \gamma, \delta$) for men and women, (ii) examining how gender interacts with other demographic and cultural variables, and (iii) investigating whether gender-based differences in PT preferences are reflected in actual financial market outcomes such as portfolio composition, trading frequency, and disposition behavior.

7. Conclusion

The study set out to consolidate the growing literature on the application of PT in finance, identify recent trends and unresolved

empirical debates, and suggest directions for future research. In that vein, we conducted a hybrid review combining SLR, bibliometric mapping, and content analysis on a corpus of 218 ABDC A*/A/B articles for the bibliometric stage and 126 ABDC A*/A articles for the content analysis, identified through the SPAR-4-SLR protocol of Paul et al. (2021). The bibliometric findings show that PT research sits at the juncture of decision science and mainstream finance. The top journals are "Journal of Economic Behavior and Organization", "Journal of Risk and Uncertainty", "Journal of Banking and Finance", and "Journal of Financial Economics". Further, the top fourteen journals account for approximately 43% of the corpus. The co-citation structure is established around Tversky and Kahneman's seminal work, while the empirical front remains inconclusive and geographically concentrated.

The content analysis reveals that the sign of the return-PTV relationship is not static and varies across markets and investor types, implying that cultural and demographic factors are important moderators. Further, no single PT component, in isolation, reconciles the major puzzles. For instance, loss aversion alone does not explain the EPP without PW (Li and Yang, 2013), and the disposition effect empirically competes with realization-utility explanations (Barberis and Xiong, 2009). Additionally, the application of PT to mutual funds, bonds, and CCs currently rests on a narrow empirical base.

Overall, the study contributes as the first hybrid review of PT in finance and the first cross-asset synthesis covering equities, mutual funds, bonds, and CCs. Also, the forward agenda in section 6.2, which follows from the content analysis's contentions, is an important contribution to advancing the field. Further, for investors and fund managers, the cross-asset evidence on the return-PTV relationship offers important implications, indicating where PT-driven mispricing is most likely to persist. For regulators, documented patterns include the managerial disposition effect (Wang et al., 2023) and intensified lottery demand during market downturns (Gong et al., 2021), which may guide disclosure design and investor-protection rules. All in all, we conclude that while the theoretical core of PT in finance is well settled, its empirical core needs further strengthening.

CRedit authorship contribution statement

Manisha Yadav: Writing – review & editing, Writing – original draft, Conceptualization.

Declaration of generative AI and AI-assisted tools in the writing process

The author (s) used Grammarly in the writing process of the manuscript, specifically for assistance in editing to improve language clarity.

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Declaration of Competing Interest

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Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at [doi:10.1016/j.jbef.2026.101204](https://doi.org/10.1016/j.jbef.2026.101204).

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